

Q5. SUDOKU PUZZLE USING BACKTRACKING

1. AIM

To design a backtracking algorithm to solve a partially filled 9×9 Sudoku puzzle by filling empty cells with valid numbers while satisfying all Sudoku constraints.

2. ALGORITHM

1. Start with the partially filled 9×9 Sudoku grid.
2. Find an empty cell in the grid.
3. Try numbers from 1 to 9 in the empty cell.
4. Check whether the number is valid in the same row, same column, and corresponding 3×3 subgrid.
5. If the number is valid, place it in the cell.
6. Recursively solve the remaining empty cells.
7. If the recursive solution fails, remove the placed number and backtrack.
8. Try the next possible number.
9. Continue until all cells are filled.
10. If no valid number can be placed, report 'No solution exists'.

3. SAMPLE INPUT

```
5 3 0 0 7 0 0 0 0
6 0 0 1 9 5 0 0 0
0 9 8 0 0 0 0 6 0
8 0 0 0 6 0 0 0 3
4 0 0 8 0 3 0 0 1
7 0 0 0 2 0 0 0 6
0 6 0 0 0 0 2 8 0
```

0 0 0 4 1 9 0 0 5
0 0 0 0 8 0 0 7 9

Note: 0 represents an empty cell.

4. SAMPLE OUTPUT

Original Sudoku:

```
5 3 0 0 7 0 0 0 0
6 0 0 1 9 5 0 0 0
0 9 8 0 0 0 0 6 0
8 0 0 0 6 0 0 0 3
4 0 0 8 0 3 0 0 1
7 0 0 0 2 0 0 0 6
0 6 0 0 0 0 2 8 0
0 0 0 4 1 9 0 0 5
0 0 0 0 8 0 0 7 9
```

Completed Sudoku:

```
5 3 4 6 7 8 9 1 2
6 7 2 1 9 5 3 4 8
1 9 8 3 4 2 5 6 7
8 5 9 7 6 1 4 2 3
4 2 6 8 5 3 7 9 1
7 1 3 9 2 4 8 5 6
9 6 1 5 3 7 2 8 4
2 8 7 4 1 9 6 3 5
3 4 5 2 8 6 1 7 9
```

5. TIME COMPLEXITY ANALYSIS

Let n be the number of empty cells. Each empty cell can have up to 9 possible values. In the worst case, the algorithm may explore all possible combinations. Therefore, the worst-case time complexity is $O(9^n)$. In practice, invalid placements are rejected early using row, column, and 3×3 subgrid checks.

Space Complexity: $O(n)$, mainly for the recursion stack and auxiliary information.

6. RESULT

The Sudoku puzzle was successfully solved using the Backtracking technique. The algorithm checks every number before placement, skips invalid placements, and backtracks whenever a selected number leads to an incorrect arrangement. Thus, it produces a valid completed Sudoku grid or reports when no solution exists.